Machine Learning 2026 • Tutorial 4 **SOLUTIONS**

Decisions and Regularization: Two Focused Problems

11 September 2026 • 08:30–09:50 • Question 1 by hand; Question 2 with a laptop

Today's plan. Question 1 starts exactly where the lecture stopped: confusion matrix, accuracy, and decision costs. Every new metric formula is provided. Question 2 applies L_2 regularization to one regression model and one classification model. The data pipeline is supplied; your work is to write the regularized fitting code, verify it, and explain its effect.

Question 1 From a confusion matrix to a decision

20 minutes

For a road-closure model, define class 1 as **impassable**. Thus a positive prediction means "close the road." At threshold $\tau = 0.5$, the model produced:

	Predicted impassable	Predicted passable	Total
Actually impassable	TP = 15	FN = 28	43
Actually passable	FP = 5	TN = 352	357
Total	20	380	$N = 400$

(a) Accuracy and the baseline. Calculate

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{N}.$$

Then calculate the accuracy of the baseline that predicts *passable* for all 400 roads. How many percentage points does the fitted model gain?

Model accuracy $= (15 + 352)/400 = 367/400 = 0.9175 = 91.75\%$. The always-passable baseline has accuracy $357/400 = 89.25\%$. The gain is $91.75 - 89.25 = 2.50$ percentage points.

(b) Use the formula sheet. Show the substituted numbers and calculate each value to four decimal places.

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}},$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \qquad \text{Specificity} = \frac{\text{TN}}{\text{TN} + \text{FP}} = 1 - \text{FPR},$$

$$F_1 = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

Precision $= 15/20 = 0.7500$; recall/TPR $= 15/43 = 0.3488$; FPR $= 5/357 = 0.0140$; specificity $= 352/357 = 0.9860$; $F_1 = 30/(30 + 5 + 28) = 30/63 = 0.4762$.

Question 1 From a confusion matrix to a decision (continued) *15 minutes*

(c) Put the error costs back into the problem. A needless closure is a false positive and costs $C_{\text{FP}} = 1$. Missing an actually impassable road is a false negative and costs $C_{\text{FN}} = 5$. For any threshold, define

$$\boxed{\text{Total cost} = 1 \cdot \text{FP} + 5 \cdot \text{FN}.}$$

The lecture compared two thresholds. Their counts are supplied:

Threshold	TP	FP	FN	TN
$\tau = 0.5$	15	5	28	352
$\tau = 1/6$	26	40	17	317

Calculate the total cost at both thresholds. Which threshold would you choose if this cost is the decision criterion? How many cost units are saved?

At $\tau = 0.5$, cost $= 1(5) + 5(28) = 145$. At $\tau = 1/6$, cost $= 1(40) + 5(17) = 125$. Choose $\tau = 1/6$; it saves 20 cost units.

(d) Explain the tradeoff. When the threshold falls from 0.5 to 1/6, complete the statements:

1. Correctly detected impassable roads increase from ____ to ____.
2. Missed impassable roads decrease from ____ to ____.
3. Needless closures increase from ____ to ____.
4. Why can the lower-accuracy rule still be the better operational decision?

TP: 15 → 26; FN: 28 → 17; FP: 5 → 40. The lower threshold creates more needless closures, but each missed impassable road costs five times as much. The weighted total therefore falls from 145 to 125, even though accuracy falls.

Connection to the lecture. For calibrated probabilities, the cost-based threshold discussed in class is

$$\tau^* = \frac{C_{\text{FP}}}{C_{\text{FP}} + C_{\text{FN}}} = \frac{1}{1+5} = \frac{1}{6}.$$

Later we will vary the threshold continuously and obtain ROC and precision–recall curves. Nothing from those future topics is required here.

Question 2 Write regularization yourself *30 minutes*

The controlled experiment. The CSV contains 400 road segments. The starter creates rainfall sensor B as an almost exact copy of sensor A, with corr(rain A, rain B) = 0.9995. The same predictors are used to predict a continuous delay and whether a road is impassable.

The starter supplies loading, scaling, a fixed 300/100 split, metrics, and reporting. **You write the model-fitting lines** using NumPy only. Place the CSV and `tutorial_04_regularization_lab.py` together, then run `python3 tutorial_04_regularization_lab.py`. Coefficient order is intercept, rain A, rain B, wetness.

(a) Build one penalty matrix. Complete `penalty_matrix(d)` so that the intercept is not penalized:

$$P = \mathrm{diag}(0, 1, 1, 1).$$

The leading zero belongs to the intercept; each 1 penalizes one slope.

Use $P = I_d$, set $P_{00} = 0$, and return P.

(b) Implement Ridge. The objectives are

$$\text{OLS: } \min_{\beta} \|y - X\beta\|_2^2, \qquad \text{Ridge: } \min_{\beta} \|y - X\beta\|_2^2 + \alpha\|\beta_{1:}\|_2^2.$$

Differentiation gives the system you must solve:

$$\boxed{(X^\top X + \alpha P)\beta = X^\top y.}$$

Complete `ridge_fit`. Use `np.linalg.solve(A,b)`; do not compute A^{-1} explicitly. Run the script and copy the output into the table.

Set $A = X^\top X + \alpha P$, $b = X^\top y$, and return `np.linalg.solve(A, b)`.

α	rain A	rain B	$\|\beta_{1:}\|_2$	train RMSE	test RMSE
0					
10					
100					

0	−1.071	7.196	7.597	2.772	3.131
10	2.953	3.071	4.761	2.777	3.107
100	2.600	2.615	4.052	2.963	3.343

(c) Read the regression experiment.

1. For $\alpha = 0$ and 10, calculate $\beta_{\text{rain A}} + \beta_{\text{rain B}}$.
2. Which fit has the more stable, balanced allocation between the two nearly identical sensors?
3. Did moderate Ridge substantially damage training or test RMSE?
4. What warning does $\alpha = 100$ give?

The sums are 6.125 for OLS and 6.024 for Ridge at $\alpha = 10$. Moderate Ridge replaces opposing coefficients $(-1.071, 7.196)$ by the balanced pair $(2.953, 3.071)$ while test RMSE changes from 3.131 to 3.107. $\alpha = 100$ shrinks more but worsens both RMSE values: stronger regularization is not automatically better.

Question 2 Logistic Ridge: add the two new terms *10 minutes*

(d) Implement Logistic Ridge. Logistic regression minimizes binary cross-entropy. Logistic Ridge adds the same preference for smaller slopes:

$$\min_{\beta} \underbrace{-\sum_i [y_i \log p_i + (1-y_i)\log(1-p_i)]}_{\text{binary cross-entropy}} + \frac{\lambda}{2}\|\beta_{1:}\|_2^2.$$

For $p = \sigma(X\beta)$ and $W = \mathrm{diag}(p_i(1-p_i))$, complete the two regularization lines in `logistic_ridge_fit`:

$$\boxed{g = X^\top(p-y) + \lambda P\beta}, \qquad \boxed{H = X^\top W X + \lambda P}.$$

The starter already computes the unregularized data gradient and Hessian. You add the two penalty terms. Its supplied line

```
step = np.linalg.solve(hessian, gradient)
```

finds s satisfying $Hs = g$, and then updates $\beta \leftarrow \beta - s$. Run the completed code and copy your output.

Use $g =$ `data_gradient + lam * penalty @ beta` and $H =$ `data_hessian + lam * penalty`.

λ	rain A	rain B	$\|\beta_{1:}\|_2$	test (TP, FP, FN, TN)
0				
1				
100				

0	1.967	−0.538	2.102	(6, 3, 9, 82)
1	0.703	0.684	1.092	(6, 3, 9, 82)
100	0.249	0.248	0.364	(0, 0, 15, 85)

(e) Compare $\lambda = 0$ with $\lambda = 1$.

1. Calculate the sum of the two rainfall coefficients in each fit.
2. What happened to the individual coefficients and their norm?
3. What happened to the test confusion matrix?
4. Using $C_{\mathrm{FP}} = 1$ and $C_{\mathrm{FN}} = 5$, calculate the test cost for both fits.

Rainfall sums are 1.429 and 1.387. With $\lambda = 1$, the pair becomes balanced and the slope norm falls from 2.102 to 1.092. The confusion matrix is unchanged, so both costs are $1(3) + 5(9) = 48$.

(f) What did your code establish? Select every supported statement.

☐ Regularization can stabilize coefficients without changing current test decisions.

☐ Regularization guarantees a lower training loss.

☐ Very large regularization can erase a useful signal.

☐ The largest possible regularization strength is always best.

☐ A validation set should be used to choose the regularization strength.

Supported: 1, 3, and 5. At $\lambda = 100$, all test roads are predicted passable: $(0, 0, 15, 85)$, with cost 75.

One conclusion across both tasks. When two predictors carry nearly the same information, the data may identify their *combined effect* much more reliably than the two individual coefficients. Moderate L_2 regularization discourages large opposing coefficients and gives a more stable allocation. Its strength must still be selected using validation performance.